Worksheet 2

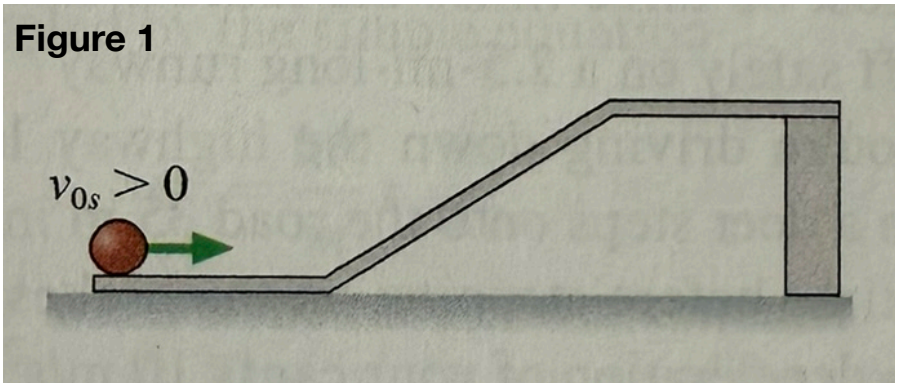
Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 9, 2026

Problems

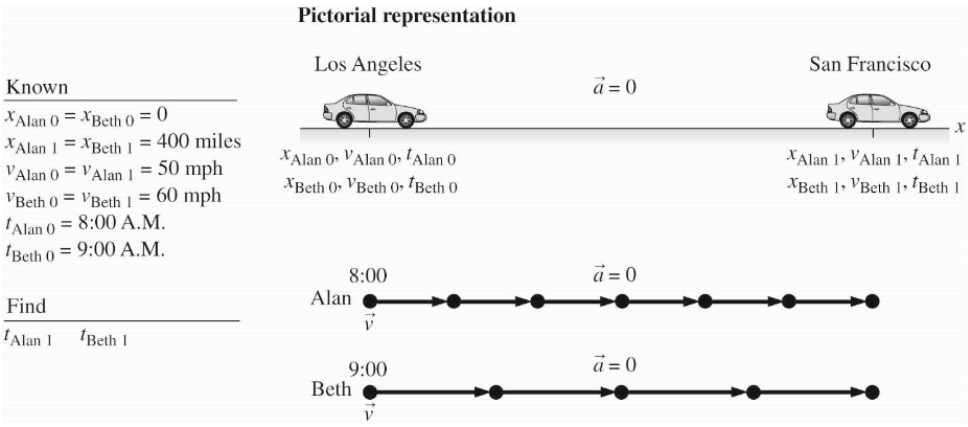
This worksheet covers motion in one-dimension, average velocity, and average acceleration.

1. Alan leaves Los Angeles at 8:00 AM to drive to San Francisco, $400\ \text{mi}$ away. He travels at a steady $50\ \text{mph}$. Beth leaves Los Angeles at 9:00 AM and drives a steady $60\ \text{mph}$
   a. Who gets to San Francisco first?
   b. How long does the first to arrive have to wait for the second?



2.3. **MODEL:** Cars will be treated by the particle model.

VISUALIZE:



SOLVE: Beth and Alan are moving at a constant speed, so we can calculate the time of arrival as follows:

$$v = \frac{\Delta x}{\Delta t} = \frac{x_1 - x_0}{t_1 - t_0} \Rightarrow t_1 = t_0 + \frac{x_1 - x_0}{v}$$

Using the known values identified in the pictorial representation, we find:

$$t_{\text{Alan 1}} = t_{\text{Alan 0}} + \frac{x_{\text{Alan 1}} - x_{\text{Alan 0}}}{v} = 8{:}00 \text{ A.M.} + \frac{400 \text{ mi}}{50 \text{ mi/h}} = 8{:}00 \text{ A.M.} + 8 \text{ h} = 4{:}00 \text{ P.M.}$$

$$t_{\text{Beth 1}} = t_{\text{Beth 0}} + \frac{x_{\text{Beth 1}} - x_{\text{Beth 0}}}{v} = 9{:}00 \text{ A.M.} + \frac{400 \text{ mi}}{60 \text{ mi/h}} = 9{:}00 \text{ A.M.} + 6.67 \text{ h} = 3{:}40 \text{ P.M.}$$

(a) Beth arrives first.

(b) Beth has to wait $t_{\text{Alan 1}} - t_{\text{Beth 1}} = 20$ min for Alan.

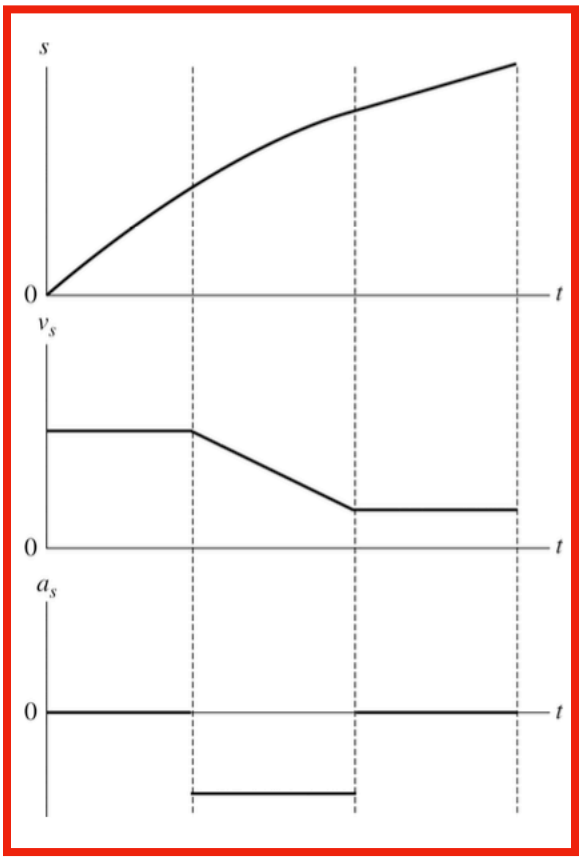
REVIEW: Times of the order of 7 or 8 h are reasonable in the present problem.

2. A ball rolls along the smooth track shown in figure 1. Each segment of the track is straight, and the ball passes smoothly from one segment to the next without changing speed or leaving the track. Draw three vertically stacked graphs showing position, velocity, and acceleration versus time. Each graph should have the same time axis and the proportions of the graph should be qualitatively correct. Assume that the ball has enough speed to reach the top.

2.43. **MODEL:** Represent the ball as a particle.

VISUALIZE: Please refer to the figure in the problem.

SOLVE: In the first and third segments the acceleration a_s is zero. In the second segment the acceleration is negative and constant. This means the velocity v_s will be constant in the first two segments and will decrease linearly in the third segment. Because the velocity is constant in the first and third segments, the position s will increase linearly. In the second segment, the position will increase parabolically rather than linearly because the velocity decreases linearly with time.



3. Calculate the average velocity of an object $(\mathrm{m/s})$ that moves from $(x, y) = (2,5)$ to $(5,9)$ in 2 seconds.

   Hint: $\Delta r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$\bar{v} = \frac{\Delta r}{\Delta t} = \frac{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}{\Delta t}$$

$$\Rightarrow \bar{v} = \frac{\sqrt{(5-2)^2 + (9-5)^2}}{2} = \frac{\sqrt{3^2 + 4^2}}{2}$$

$$\Rightarrow \bar{v} = 2.5 \ \frac{m}{s}$$

4. A particle moves along the the x-axis according to the relation $x = 6 - 6t + t^2$, where x is measured in meters and t is measured in seconds.

   a. Find the values of x for $t = 1, 2, 3, 4, 5$

   b. Find the values of velocity v for $t = 1, 2, 3, 4, 5$

   c. For each value of t indicate whether the particle is moving toward an increasing or decreasing x.

   d. Is there ever an instant when the velocity is zero?

   e. Is there a time after $t = 5$ when the particle is moving toward decreasing x?

(8) (a) For $t = 1, 2, 3, 4,$ and 5 s we have $x = 1, -2, -3, -2,$ and 1 m, (b) For $t = 1, 2, 3, 4,$ and 5 s we have $v = -4, -2, 0, 2,$ and 4 m/s, (c) For $t = 1, 2, 3, 4,$ and 5 s we have: motion toward decreasing x, motion toward decreasing x, momentarily no motion, motion toward increasing x, and motion toward increasing x, (d) Yes, at $t = 3$ s, (e) No